

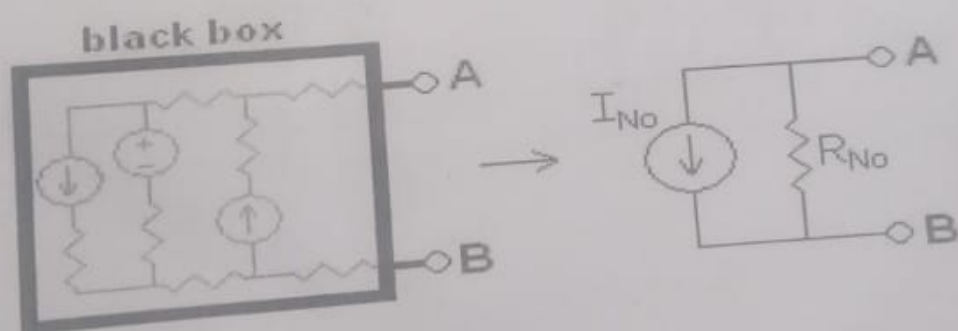
Aim: To study & verify Norton's theorem.

Apparatus Required:

1. Trainer kit
2. Connecting cords

Theorem:

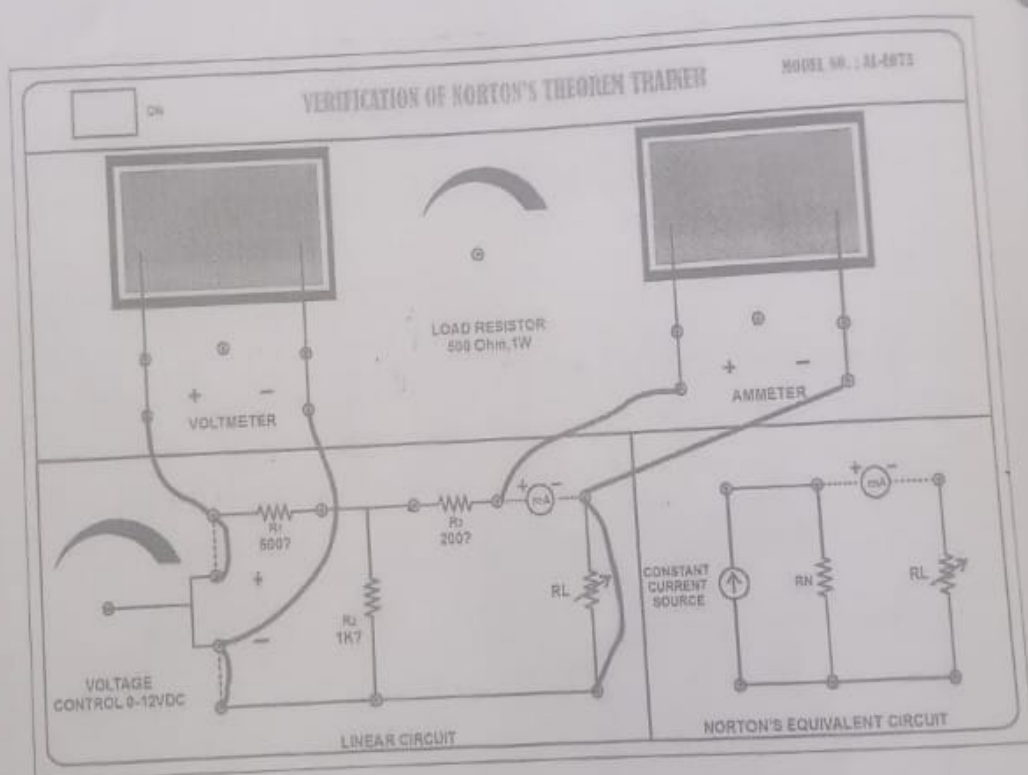
Norton's theorem states that any two terminal linear network with current sources, voltage sources and resistances can be replaced by an equivalent circuit consisting of a current source in parallel with a resistance. The value of the current source is the short circuit current between the two terminals of the network and the resistance is equal to the equivalent resistance measured between the terminals with all the energy sources are replaced by their internal resistances.



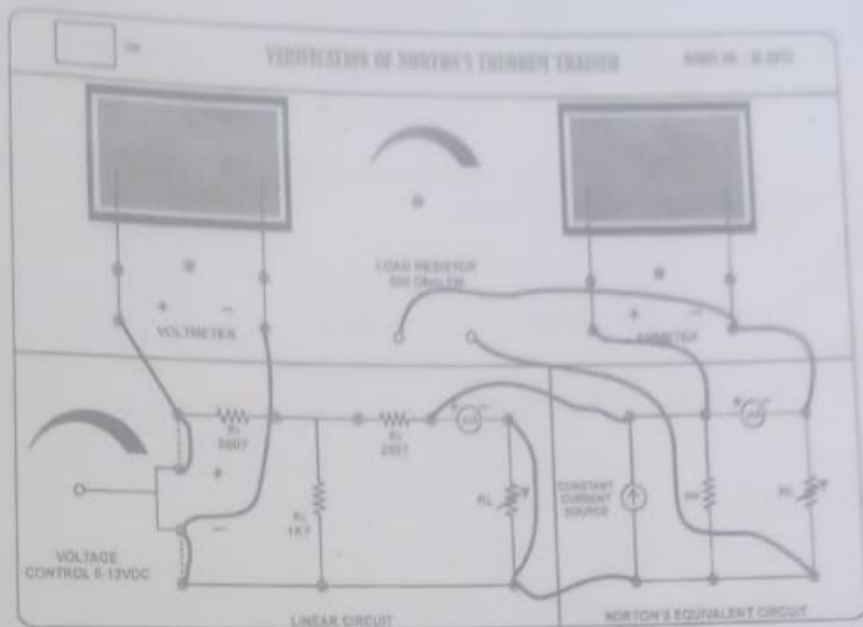
Any black box containing resistances only and voltage and current sources can be replaced by an equivalent circuit consisting of an equivalent current source in parallel connection with an equivalent resistance.

Connection diagram:

1. For calculation of short circuit current (Norton current)



2. For calculation of load current



Procedure:

1. Connection should be as per the circuit diagram.
2. Vary the supply voltage and take the corresponding reading from the ammeter.
3. Remove the load resistance & short that terminal.
4. Find the Norton's resistance R_N i.e equivalent resistance of the circuit.
5. Find the Norton's current I_N or short circuit current.
6. Set constant current source value equals to the Norton's current which is controlled by variable voltage.

7. Now connect the load and measure the current in the load resistance directly.

8. Using formula find out the current in the load resistance and Norton's current.

Observations and calculation:

Sr no.	Vin	IN (Norton current)		Req (equivalent resistance)	IL (Load current)	
		Practical	Theoretical		Practical	Theoretical
1.						
2.						
3.						
4.						
5.						

Result: If Practical and theoretical values meet near approximation. Then required result is verified.